Introduction to Category Theory

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Categories

Definition

A *category* $\mathcal{C}$ consists of the following data:

- A collection $\mathrm{Ob}(\mathcal{C})$ of *objects*;
- For each pair of objects A, B, a collection $\mathrm{hom}_{\mathcal{C}}(A, B)$ whose elements are called *morphisms* or *arrows*;
- Composition: Given two morphisms $f : A \to B$ e $g : B \to C$, there exists a morphism $g \circ f : A \to C$;

Such that:

- Composition is associative;
- For each object A, there exists a morphism $1_A : A \to A$ called the *identity morphism* that acts as an identity for composition;

Notation:

We write $A \in \mathcal{C}$ for $A \in \mathrm{Ob}(\mathcal{C})$
$\mathrm{hom}_{\mathcal{C}}(A, B)$, $\mathrm{hom}(A, B)$ e $\mathcal{C}(A, B)$

Examples

- $\mathbf{Set}$, $\mathbf{Grp}$, $\mathbf{Mon}$, $\mathbf{Ring}$, $\mathbf{Vect}_{\mathbb{R}}$, $\mathbf{Top}$
- $\mathbf{N}$
- Preorder / Partial order $(X, \leq)$
- $\mathbf{Mat}(\mathbb{R})$
- Monoid M / group G

Types of Morphisms

Definition

Let $\mathcal{C}$ be a category. A morphism $f : A \to B$ is said to be an *isomorphism* or *invertible*, if there exists a morphism $g : B \to A$ such that $gf = 1_A$ e $fg = 1_B$.
If that is the case, we say that the objects A and B are *isomorphic*, which we denote by $A \cong B$.

Note: if f is an isomorphism, then its inverse is unique, so we use the notation f^{-1}.

Definition

A group G is a category with only one object where every morphism is an isomorphism.

Note: A category where every morphism is an isomorphism is called a *groupoid*.

Monomorphisms

Proposition

Let X, Y be sets and $f : X \to Y$ a function. The following are equivalent:

- *f is injective;*
- *Given functions g, h:*

$$
\begin{tikzcd}
Z \arrow[r, shift left, "g"] \arrow[r, shift right, "h"'] & X \arrow[r, "f"] & Y
\end{tikzcd}
$$

  If $fg = fh$, then $g = h$;
- *There exists a function $w : Y \to X$ such that $wf = 1_X$.*

Definition

Let $\mathcal{C}$ be a category. A morphism $f : A \to B$ is a *monomorphism* or *mono* if, for any morphisms $g, h : C \rightrightarrows A$, the equality $fg = fh$ implies that $g = h$.

Epimorphisms

Proposition

Let X, Y be sets and $f : X \to Y$ a function. The following are equivalent:

- *f is surjective;*
- *Given two functions g, h:*

$$X \xrightarrow{f} Y \overset{g}{\underset{h}{\rightrightarrows}} Z$$

If $gf = hf$, then $g = h$;

- *There exists a function $w : Y \to X$ such that $fw = 1_Y$.*

Definition

Let $\mathcal{C}$ be a category. A morphism $f : A \to B$ is an *epimorphism* or *epic* if, for any morphisms $g, h : B \rightrightarrows C$, $gf = hf$ implies $g = h$.

Iso, Mono e Epi

Proposition

Let $\mathcal{C}$ be a category. If f is an isomorphism, then f is both a monomorphism and an epimorphism.

Note: f being a monomorphism $+$ epimorphism does not imply that f is an isomorphism.
A category where this happens is called *balanced*.

Terminal and initial objects

Definition

Let $\mathcal{C}$ be a category. An object A is called:

- *terminal* if for every object B there exists exactly one morphism $B \to A$;
- *initial* if for every object B there exists exactly one morphism $A \to B$.

Proposition

Let $\mathcal{C}$ be a category. Then:

- *If $\mathcal{C}$ has a terminal object, then it is unique up to isomorphism;*
- *If $\mathcal{C}$ has an initial object, then it is unique up to isomorphism;*

Note: the terminal object is usually denoted by 1 and the initial object by 0 or $\emptyset$.

Opposite Category

Definition

Let $\mathcal{C}$ be a category. The *opposite category* of $\mathcal{C}$, denoted by $\mathcal{C}^{\mathrm{op}}$, is given by:

- The objects of $\mathcal{C}^{\mathrm{op}}$ are the objects of $\mathcal{C}$;
- The morphisms $\mathcal{C}^{\mathrm{op}}$ are the same as the morphisms in $\mathcal{C}$, but we 'flip' the direction. Formally:

$$\mathsf{hom}_{\mathcal{C}^{\mathrm{op}}}(A, B) := \mathsf{hom}_{\mathcal{C}}(B, A)$$

- Composition is given by $g \circ_{\mathrm{op}} f = f \circ g$;

Proposition

Let $\mathcal{C}$ be a category. Then:

- *An object A is terminal [initial] em $\mathcal{C}$ iff A is initial [terminal] in $\mathcal{C}^{\mathrm{op}}$;*
- *A morphism f is mono [epic] in $\mathcal{C}$ iff f is epic [mono] in $\mathcal{C}^{\mathrm{op}}$.*

Product of Categories

Definition

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. We define $\mathcal{C} \times \mathcal{D}$ as being the category given by:

- The objects are pairs (A, B), where $A \in \mathcal{C}$ e $B \in \mathcal{D}$. In other words $\mathrm{Ob}(\mathcal{C} \times \mathcal{D}) = \mathrm{Ob}(\mathcal{C}) \times \mathrm{Ob}(\mathcal{D})$;
- A morphism $(A, B) \to (A', B')$ is given by a pair (f, g) where $f : A \to A'$ is a morphism in $\mathcal{C}$ and $g : B \to B'$ is a morphism in $\mathcal{D}$;

Composition of two morphisms:

$$(A, B) \xrightarrow{(f,g)} (A', B') \xrightarrow{(f',g')} (A'', B'')$$

is defined as $(f', g') \circ (f, g) = (f' \circ f, g' \circ g)$.

Functors

Definition

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A *functor* $F : \mathcal{C} \to \mathcal{D}$ is given by:

- For each object $A \in \mathcal{C}$, an object $F(A)$ in $\mathcal{D}$;
- Given a morphism $f : A \to B$ em $\mathcal{C}$, a morphism $F(f)$ in $\mathcal{D}$:

$$\begin{array}{ccc} A & & F(A) \\ {\scriptstyle f}\downarrow & \stackrel{F}{\longmapsto} & \downarrow{\scriptstyle F(f)} \\ B & & F(B) \end{array}$$

Such that $F(1_A) = 1_{F(A)}$ for every object $A \in \mathcal{C}$, and $F(f \circ g) = F(f) \circ F(g)$ for all composable morphisms $A \xrightarrow{g} B \xrightarrow{f} C$ in $\mathcal{C}$.

Functors II

Definition

Let $\mathcal{C} \xrightarrow{F} \mathcal{D} \xrightarrow{G} \mathcal{K}$ be functors. We define the composition $G \circ F : \mathcal{C} \to \mathcal{K}$ to be the functor given by:

- For any $A \in \mathcal{C}$, then $(G \circ F)(A) = G(F(A))$;
- For any morphism f in $\mathcal{C}$, then $(G \circ F)(f) = G(F(f))$.

Definition

We define **Cat** as being the category whose objects are categories, and morphisms are functors, where composition is defined as above.

Definition

Two categories $\mathcal{C}$ and $\mathcal{D}$ are said to be isomorphic if they are isomorphic as objects in **Cat**. In other words, if there are functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ such that $GF = 1_{\mathcal{C}}$ e $FG = 1_{\mathcal{D}}$.

Full and Faithful

Note: Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. For each pair of objects $A, B \in \mathcal{C}$, the functor F induces a map

$$\begin{aligned} F : \hom_{\mathcal{C}}(A, B) &\to \hom_{\mathcal{D}}(F(A), F(B)) \\ f &\mapsto F(f) \end{aligned} \tag{1}$$

Definition

A functor $F : \mathcal{C} \to \mathcal{D}$ is said to be:

- *Full*, if the map (1) is surjective;
- *Faithful*, if the map (1) is injective.

Proposition

Let $F : \mathcal{C} \to D$ be a functor. Then F is an isomorphism iff it is bijective on objects, and fully faithful.

Functors III

Proposition

Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. Then:

1. *F preserves isomorphisms;*
2. *If F is fully faithful, then F reflects isomorphisms;*
3. *If F is faithful, then F reflects monomorphisms and epimorphisms.*

Corollary

Let $F : \mathcal{C} \to \mathcal{D}$ be a functor and A, B objects in $\mathcal{C}$. If $A \cong B$, then $F(A) \cong F(B)$, where the converse is true if, for example, F is fully faithful.

Contravariant Functors

Definition

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A *contravariant functor* $F : \mathcal{C} \to \mathcal{D}$ is given by:

- For each object $A \in \mathcal{C}$, an object $F(A)$ in $\mathcal{D}$;
- Given a morphism $f : A \to B$ em $\mathcal{C}$, a morphism $F(f)$ in $\mathcal{D}$:

$$\begin{tikzcd} A \arrow[d, "f"'] & & F(A) \\ {} \arrow[rr, phantom, "\overset{F}{\longmapsto}"] & & {} \\ B & & F(B) \arrow[u, "F(f)"'] \end{tikzcd}$$

Such that $F(1_A) = 1_{F(A)}$ for every object $A \in \mathcal{C}$, and $F(f \circ g) = F(g) \circ F(f)$ for all composable morphisms $A \xrightarrow{g} B \xrightarrow{f} C$ in $\mathcal{C}$.

Contravariant Functors II

Proposition

A contravariant functor from $\mathcal{C}$ to $\mathcal{D}$ is the same thing as a functor $F : \mathcal{C}^{\mathrm{op}} \to \mathcal{D}$.

Proposition

Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. Then F is also a functor from $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}^{\mathrm{op}}$.

Note: As such, a functor $F : \mathcal{C}^{\mathrm{op}} \to \mathcal{D}$ is the same thing as a functor $F : \mathcal{C} \to \mathcal{D}^{\mathrm{op}}$.

Proposition

The assignment $\mathcal{C} \mapsto \mathcal{C}^{\mathrm{op}}$ and $F \mapsto F$ defines a functor $\mathbf{Cat} \to \mathbf{Cat}$.

Natural Transformations

Definition

Let $F, G : \mathcal{C} \rightrightarrows \mathcal{D}$ be two functors. A *natural transformation* $\alpha : F \Rightarrow G$ is given by, for each $C \in \mathcal{D}$, a morphism

$$\alpha_C : F(C) \to G(C)$$

in $\mathcal{D}$, called the *components* of α in C, such that for each morphism $f : A \to B$ in $\mathcal{C}$, the following diagram called the *naturality square* of α commutes:

$$\begin{tikzcd} F(A) \arrow[r, "\alpha_A"] \arrow[d, "F(f)"'] & G(A) \arrow[d, "G(f)"] \\ F(B) \arrow[r, "\alpha_B"'] & G(B) \end{tikzcd}$$

Composition of Natural Transformations

Definition

Let $\mathcal{C}$ and $\mathcal{D}$ be two categories, and consider the following functors and natural transformations:

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left=50, "F"{name=F}] \arrow[r, "G"{description, name=G}] \arrow[r, bend right=50, "H"'{name=H}] & \mathcal{D} \arrow[from=F, to=G, Rightarrow, "\alpha"] \arrow[from=G, to=H, Rightarrow, "\beta"] \end{tikzcd}$$

We define the composite $\beta \circ \alpha : F \Rightarrow H$ as the natural transformation with components given by the composition of α_A followed by β_A:

$$(\beta \circ \alpha)_A : F(A) \xrightarrow{\alpha_A} G(A) \xrightarrow{\beta_A} H(A)$$

so $(\beta \circ \alpha)_A = \beta_A \circ \alpha_A$ for each $A \in \mathcal{C}$.

Functor Categories

Definition

Let $\mathcal{C}$ and $\mathcal{D}$ be two categories. We define $[\mathcal{C}, \mathcal{D}]$ as being the category whose objects are functors from $\mathcal{C}$ to $\mathcal{D}$ and morphisms are natural transformations.

Definition

Two functors $F, G : \mathcal{C} \rightrightarrows \mathcal{D}$ are said to be *naturally isomorphic*, written as $F \cong G$, if they are isomorphic as objects in $[\mathcal{C}, \mathcal{D}]$, i.e. is there are natural transformations $\alpha : F \Rightarrow G$ and $\beta : G \Rightarrow F$ such that $\alpha \circ \beta = 1_G$ and $\beta \circ \alpha = 1_F$. If this is the case, we say that α and β are *natural isomorphisms*.

Proposition

Let $F, G : \mathcal{C} \rightrightarrows \mathcal{D}$ be two functors and $\alpha : F \Rightarrow G$ a natural transformation. Then α is a natural isomorphism iff for each $A \in \mathcal{C}$, the component $\alpha_A : F(A) \to G(A)$ is an isomorphism in $\mathcal{D}$.

Equivalent Categories

Note: Given two functors $F, G : \mathcal{C} \to \mathcal{D}$, we say that

$$F(A) \cong G(A), \text{ naturally in } A$$

to mean that F and G are naturally isomorphic.

Definition

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. We say that $\mathcal{C}$ and $\mathcal{D}$ are *equivalent*, denoted by $\mathcal{C} \simeq \mathcal{D}$, if there are functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ such that $F \circ G \cong 1_{\mathcal{D}}$ and $G \circ F \cong 1_{\mathcal{C}}$. If this is the case, then we say that F is an *equivalence* of categories.

A functor $F : \mathcal{C} \to \mathcal{D}$ is said to be *essentially surjective on objects*, if for each $B \in \mathcal{D}$, there is some $A \in \mathcal{C}$ such that $F(A) \cong B$.

Proposition

Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. Then F is an equivalence iff F is fully faithful and essentially surjective on objects.

SubCategories

Definition

Let $\mathcal{C}$ be a category. A *subcategory* $\mathcal{D}$ is given by:

- A subset $\mathrm{Ob}(\mathcal{D}) \subseteq \mathrm{Ob}(\mathcal{C})$ of objects;
- For each pair $A, B \in \mathrm{Ob}(\mathcal{D})$, a subset of morphisms $\hom_{\mathcal{D}}(A, B) \subseteq \hom_{\mathcal{C}}(A, B)$

Such that $\mathcal{D}$ is itself a category.

Definition

A subcategory $\mathcal{D} \subseteq \mathcal{C}$ is said to be a *full subcategory* if $\hom_{\mathcal{D}}(A, B) = \hom_{\mathcal{C}}(A, B)$ for all $A, B \in \mathcal{D}$, i.e. if the inclusion functor $\iota : \mathcal{D} \hookrightarrow \mathcal{C}$ is full. (note that the inclusion functor is always faithful).

SubCategories II

Definition

A category $\mathcal{C}$ is said to be *skeletal*, if for any two objects $A, B \in \mathcal{C}$, if $A \cong B$ then $A = B$.
A *skeleton* of a category $\mathcal{C}$ is a skeletal subcategory whose inclusion functor is an equivalence.

Theorem

Every category has a skeleton. Furthermore, such skeleton is unique up to isomorphism.
Furthermore, two categories are equivalent if and only if they have isomorphic skeletons.

Dualities

An equivalence $\mathcal{C}^{\mathrm{op}} \simeq \mathcal{D}$ is called a *duality* between the categories $\mathcal{C}$ and $\mathcal{D}$.

Examples:

- Stone duality:
$$\{\text{Boolean Algebras}\}^{\mathrm{op}} \simeq \{\text{totally disconnected compact } T_2 \text{ spaces}\}$$
- Gelfand-Naimark duality:
$$\{\text{Commutative unital } C^*\text{-algebras}\}^{\mathrm{op}} \simeq \{\text{compact } T_2 \text{ spaces}\}$$
- In algebraic geometry, given a ACF k:
$$\{\text{Affine Varieties over } k\}^{\mathrm{op}} \simeq \{\text{f.g. } k\text{-algebras with no nontrivial nilpotents}\}$$
- Pontryagin Duality:
$$\{\text{Locally compact abelian top. Groups}\}^{\mathrm{op}} \simeq \{\text{Locally compact abelian top. Groups}\}$$

Adjunctions

Definition

Consider categories and functors as follows:

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "L"] & \mathcal{D} \arrow[l, bend left, "R"] \end{tikzcd}$$

We say that L is *left adjoint* to R and that R is *right adjoint* to L, written as $L \dashv R$, if there exists an isomorphism

$$\hom_{\mathcal{D}}(L(A), B) \cong \hom_{\mathcal{C}}(A, R(B)) \tag{2}$$

natural in $A \in \mathcal{C}$ and $B \in \mathcal{D}$.
An *adjunction* between L and R is a choice of isomorphism as in (2).

Adjunctions II

The bijection $\hom_{\mathcal{D}}(L(A), B) \cong \hom_{\mathcal{C}}(A, R(B))$ creates the assignment:

$$\left(L(A) \xrightarrow{g} B\right) \mapsto \left(A \xrightarrow{\bar{g}} R(B)\right)$$
$$\left(L(A) \xrightarrow{\bar{f}} B\right) \leftarrow\!\shortmid \left(A \xrightarrow{f} R(B)\right)$$

Where $\bar{\bar{f}} = f$. We call $\bar{f}$ the *transpose* of f. Naturality is the same thing as the following two conditions

$$\overline{\left(L(A) \xrightarrow{g} B \xrightarrow{q} B'\right)} = \left(A \xrightarrow{\bar{g}} R(B) \xrightarrow{R(q)} R(B')\right) \tag{3}$$
$$\overline{\left(A' \xrightarrow{p} A \xrightarrow{f} R(B)\right)} = \left(L(A') \xrightarrow{L(p)} L(A) \xrightarrow{\bar{f}} B\right) \tag{4}$$

That is, $\overline{q \circ g} = R(q) \circ \bar{g}$ and $\overline{f \circ p} = \bar{f} \circ L(p)$

Adjunctions III

Proposition

Consider adjunctions

$$
\begin{tikzcd}
\mathcal{C} \arrow[r, bend left, "L"] \arrow[r, phantom, "\perp"] & \mathcal{D} \arrow[l, bend left, "R"] \arrow[r, bend left, "L'"] \arrow[r, phantom, "\perp"] & \mathcal{E} \arrow[l, bend left, "R'"]
\end{tikzcd}
$$

Then, $L' \circ L \dashv R \circ R'$. *In fact, we have*

$$\hom_{\mathcal{E}}(L'L(A), B) \cong \hom_{\mathcal{D}}(L(A), R'(B)) \cong \hom_{\mathcal{C}}(A, RR'(B))$$

Proposition

Left adjoints preserve initial objects and right adjoints preserve terminal objects.

Unit and Counit

Consider an adjunction

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "L"] \arrow[r, phantom, "\perp"] & \mathcal{D} \arrow[l, bend left, "R"] \end{tikzcd}$$

From the bijection $\hom_{\mathcal{D}}(L(A), B) \cong \hom_{\mathcal{C}}(A, R(B))$, we obtain the following bijection natural in $A \in \mathcal{C}$:

$$\hom_{\mathcal{D}}(L(A), L(A)) \cong \hom_{\mathcal{C}}(A, RL(A)) \tag{5}$$

In particular, for each $A \in \mathcal{C}$, we have a morphism

$$\left(A \xrightarrow{\eta_A} RL(A)\right) = \overline{\left(L(A) \xrightarrow{1} L(A)\right)}$$

Proposition

Consider an adjunction as above. Then, the maps $\eta_A : A \to RL(A)$ form a natural transformation $\eta : 1_{\mathcal{C}} \Rightarrow RL$, known as the unit of the adjunction $L \dashv R$.

Unit and Counit II

Consider an adjunction

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "L"] \arrow[r, phantom, "\perp"] & \mathcal{D} \arrow[l, bend left, "R"] \end{tikzcd}$$

From the bijection $\hom_{\mathcal{D}}(L(A), B) \cong \hom_{\mathcal{C}}(A, R(B))$, we obtain the following bijection natural in $B \in \mathcal{D}$:

$$\hom_{\mathcal{D}}(LR(B), B) \cong \hom_{\mathcal{C}}(R(B), R(B)) \tag{6}$$

In particular, for each $B \in \mathcal{D}$, we have a morphism

$$\left(LR(B) \xrightarrow{\varepsilon_B} B\right) = \overline{\left(R(B) \xrightarrow{1} R(B)\right)}$$

Proposition

Consider an adjunction as above. Then, the maps $\varepsilon_B : LR(B) \to B$ form a natural transformation $\varepsilon : LR \Rightarrow 1_{\mathcal{D}}$, known as the counit of the adjunction $L \dashv R$.

Whiskering of natural transformations

Consider the following:

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "F"] \arrow[r, bend right, "H"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{D} \arrow[r, "G"] & \mathcal{E} \end{tikzcd}$$

Then we have a composite:

$$\begin{tikzcd} \mathcal{C} \arrow[rr, bend left, "GF"] \arrow[rr, bend right, "GH"'] \arrow[rr, phantom, "\Downarrow G\alpha"] & & \mathcal{E} \end{tikzcd}$$

With components

$$(G\alpha)_A := G(\alpha_A) : G(F(A)) \to G(H(A))$$

Whiskering of natural transformations II

Consider the following:

$$\begin{tikzcd} \mathcal{E} \arrow[r, "G"] & \mathcal{C} \arrow[r, bend left, "F"] \arrow[r, bend right, "H"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{D} \end{tikzcd}$$

Then we have a composite:

$$\begin{tikzcd} \mathcal{E} \arrow[r, bend left, "FG"] \arrow[r, bend right, "HG"'] \arrow[r, phantom, "\Downarrow\alpha G"] & \mathcal{D} \end{tikzcd}$$

With components

$$(\alpha G)_A := \alpha_{G(A)} : F(G(A)) \to H(G(A))$$

Unit and Counit III

Proposition

Consider an adjunction

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "L"] \arrow[r, phantom, "\perp"] & \mathcal{D} \arrow[l, bend left, "R"] \end{tikzcd}$$

with unit η and counit ε. Then the following triangles commute (in $[\mathcal{C}, \mathcal{D}]$ and $[\mathcal{D}, \mathcal{C}]$ resp.)

$$\begin{tikzcd} L \arrow[r, "L\eta"] \arrow[dr, "1_L"'] & LRL \arrow[d, "\varepsilon L"] \\ & L \end{tikzcd} \qquad \begin{tikzcd} R \arrow[r, "\eta R"] \arrow[dr, "1_R"'] & RLR \arrow[d, "R\varepsilon"] \\ & R \end{tikzcd}$$

These two triangles are known as the *triangle identities*.

Unit and Counit IV

Proposition

Consider an adjunction

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "L"] \arrow[r, phantom, "\perp"] & \mathcal{D} \arrow[l, bend left, "R"] \end{tikzcd}$$

with unit η *and counit* ε*, and let* $A \in \mathcal{C}$ *and* $B \in \mathcal{D}$*. Then, for any* $g : L(A) \to B$,

$$\bar{g} = R(g) \circ \eta_A$$

and for any $g : A \to R(B)$*, we have that*

$$\bar{f} = \varepsilon_B \circ L(f)$$

Adjunctions = Unit and Counit

Theorem

Consider two categories and functors

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "L"] & \mathcal{D} \arrow[l, bend left, "R"] \end{tikzcd}$$

Then, there is a one-to-one correspondence between:

1. *Adjunctions between L and R (with L being left adjoint);*
2. *Pairs $(\eta : 1_{\mathcal{C}} \Rightarrow RL, \varepsilon : LR \Rightarrow 1_{\mathcal{D}})$ of natural transformations satisfying the triangle identities.*

Corollary

Using the same notation as the last theorem, there exists an adjunction $L \dashv R$ if and only if there are natural transformations $\eta : 1_{\mathcal{C}} \Rightarrow RL$ and $\varepsilon : LR \Rightarrow 1_{\mathcal{D}}$ satisfying the triangle identities.

Adjunctions IV

So we can think of adjunctions as:

1. Triples (L, R, Φ) with

$$\hom_{\mathcal{D}}(L(-), -) \overset{\Phi}{\cong} \hom_{\mathcal{C}}(-, R(-))$$

2. Quadruples $(L, R, \eta, \varepsilon)$ where $\eta : 1_{\mathcal{C}} \Rightarrow RL$ and $\varepsilon : LR \Rightarrow 1_{\mathcal{D}}$ satisfying the triangle identities.

Sizes of Categories

Definition

A category $\mathcal{C}$ is said to be:

- *Small*, if the collection of all arrows is a set;
- *Locally small* if for every $A, B \in \mathcal{C}$, the collection $\hom_{\mathcal{C}}(A, B)$ is a set;
- *Essentially small* if it is equivalent to a small category;

Representables I

Definition

Let $\mathcal{C}$ be a locally small category, and fix an object A. We define a functor

$$h^A : \mathcal{C} \to \mathbf{Set}$$

as follows:

- For $B \in \mathcal{C}$, we define $h^A(B) = \hom(A, B)$;
- For $f : B \to C$, then $h^A(f)$ is given by

$$\begin{aligned} h^A(f) : h^A(B) &\to h^A(C) \\ h &\mapsto f \circ h \end{aligned}$$

The notation $\hom(A, -)$ is also common for h^A. We also sometimes denote $h^A(f)$ by $f \circ -$ or f_*.

Representables II

Definition

Let $\mathcal{C}$ be a locally small category and $F : \mathcal{C} \to \mathbf{Set}$ a functor. We say that F is *representable* if there exists some object $A \in \mathcal{C}$ such that $F \cong h^A$. A *representation* of F is a choice of object A and isomorphism between h^A and F.

Representables functors are sometimes just called *representables*.

Theorem

Any functor $R : \mathcal{C} \to \mathbf{Set}$ *with a left adjoint is representable.*

Representables III

Let $\mathcal{C}$ be a locally small category and consider a morphism $f : A \to B$. Then this induces a natural transformation:

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left=50, "h^B"{name=U}] \arrow[r, bend right=50, "h^A"'{name=D}] & \mathbf{Set} \arrow[Rightarrow, from=U, to=D, "h^f"] \end{tikzcd}$$

Where, for each $C \in \mathcal{C}$, the component of h^f is

$$\begin{aligned} (h^f)_C : h^B(C) &\to h^A(C) \\ p &\mapsto p \circ f \end{aligned}$$

Representables IV

Definition

Let $\mathcal{C}$ be a locally small category. We define a functor

$$h^{\bullet} : \mathcal{C}^{\mathrm{op}} \to [\mathcal{C}, \mathbf{Set}]$$

Where, given an object A, its image is h^A and, given an arrow f, its image is h^f.

Representables V

Definition

Let $\mathcal{C}$ be a locally small category, and fix an object A. We define a functor

$$h_A : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$$

as follows:

- For $B \in \mathcal{C}$, we define $h_A(B) = \mathrm{hom}(B, A)$;
- For $f : B \to C$, then $h_A(f)$ is given by

$$\begin{aligned} h_A(f) : h_A(C) &\to h_A(B) \\ h &\mapsto h \circ f \end{aligned}$$

The notation $\mathrm{hom}(-, A)$ is also common for h_A. We also sometimes denote $h_A(f)$ by $- \circ f$ or f^*.

Representables VI

Definition

Let $\mathcal{C}$ be a locally small category and $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ a functor. We say that F is *representable* if there exists some object $A \in \mathcal{C}$ such that $F \cong h_A$. A *representation* of F is a choice of object A and isomorphism between h_A and F.

Recall: Functors of the form $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ are usually called *presheaves* on $\mathcal{C}$.

Representables VII

Let $\mathcal{C}$ be a locally small category and consider a morphism $f : A \to B$. Then this induces a natural transformation:

$$\begin{tikzcd} \mathcal{C}^{\mathrm{op}} \arrow[r, bend left=50, "h_A"{name=U}] \arrow[r, bend right=50, "h_B"'{name=D}] & \mathbf{Set} \arrow[from=U, to=D, Rightarrow, "h_f"] \end{tikzcd}$$

Where, for each $C \in \mathcal{C}$, the component of h_f is

$$\begin{aligned} (h_f)_C : h_A(C) &\to h_B(C) \\ p &\mapsto f \circ p \end{aligned}$$

Representables VIII

Definition

Let $\mathcal{C}$ be a locally small category. The *Yoneda embedding* is the functor

$$h_\bullet : \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$$

Where, given an object A, its image is h_A and, given an arrow f, its image is h_f.

Yoneda Lemma

Theorem (Yoneda)

Let $\mathcal{C}$ be a locally small category. Then

$$\hom_{[\mathcal{C}^{\mathrm{op}},\mathbf{Set}]}(h_A, F) \cong F(A)$$

naturally in $A \in \mathcal{C}$ and $F \in [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$.

Corollary

For any locally small category $\mathcal{C}$, the Yoneda embedding

$$h_\bullet : \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$$

is fully faithful.

Yoneda Lemma II

Corollary

Let $\mathcal{C}$ be a locally small category, and $A, B \in \mathcal{C}$. Then:

$$h_A \cong h_B \iff A \cong B \iff h^A \cong h^B$$

Corollary

Consider functors

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left=60, "L'"] \arrow[r, bend left=20, "L"'] & \mathcal{D} \arrow[l, bend left=30, "R"] \end{tikzcd}$$

If $L \dashv R$ and $L' \dashv R$, then $L \cong L'$, i.e. the left adjoint of a fixed functor R are unique, up the isomorphism. The same goes for right adjoints.

Products

Definition

Let $\mathcal{C}$ be a category and consider objects A, B. A *product* of A and B (if it exists) is an object P and arrows $\pi_A : P \to A$ and $\pi_B : P \to B$ with the property that, given any arrows $f : C \to A$ and $g : C \to B$, then there exists a unique arrow $(f, g) : C \to P$ such that the following commutes:

$$\begin{tikzcd} & C \arrow[dl, "f"'] \arrow[d, dashed, "{(f,g)}" description] \arrow[dr, "g"] & \\ A & P \arrow[l, "\pi_A"] \arrow[r, "\pi_B"'] & B \end{tikzcd}$$

Note: We will later see that the product of A and B, if it exists, is unique up to isomorphism, and so we use the notation $A \times B$.

Equalizers

Definition

Consider two arrows $f, g : A \to B$. An equalizer of f and g (if it exists) is an object E with a map $E \xrightarrow{i} A$ such that $fi = gi$ with the property that for any other $X \xrightarrow{w} A$ with $fw = gw$, there exists a unique map $X \xrightarrow{\bar{w}} E$ such that the following commutes

$$\begin{tikzcd} E \arrow[r, "i"] & A \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B \\ X \arrow[u, dashed, "\bar{w}"] \arrow[ur, "w"'] & & \end{tikzcd}$$

Note: We will see later that the equalizer of a pair of parallel morphisms is unique, up to isomorphism.

Lemma

Every equalizer is a monomorphism. (A monomorphism which is the equalizer of some pair of parallel arrows is called a regular monomorphism)

Pullbacks I

Definition

Let $\mathcal{C}$ be a category and consider two morphisms $f : A \to Z$ and $g : B \to Z$. A pullback of f and g (if it exists), is an object P with maps $P \xrightarrow{p_1} A$ and $P \xrightarrow{p_2} B$ such that $fp_1 = gp_2$, with the property that, for any other object W with maps $W \xrightarrow{w_1} A$ and $W \xrightarrow{w_2} B$ such that $fw_1 = gw_2$, there exists a unique map $W \to P$ such that the following commutes:

$$
\begin{tikzcd}
W \arrow[dr, dashed] \arrow[drr, bend left, "w_2"] \arrow[ddr, bend right, "w_1"'] & & \\
& P \arrow[r, "p_2"] \arrow[d, "p_1"'] & B \arrow[d, "g"] \\
& A \arrow[r, "f"'] & Z
\end{tikzcd}
$$

Pullbacks II

Note 1: We will see later that the pullback of two morphisms, if it exists, is unique up to isomorphism. As such, the pullback of $f : A \to Z$ and $g : B \to Z$ is sometimes denoted by $A \times_Z B$.

Note 2: The a commutative square

$$\begin{tikzcd} P \arrow[r, "p_2"] \arrow[d, "p_1"'] \arrow[dr, phantom, very near start, "\lrcorner"] & B \arrow[d, "g"] \\ A \arrow[r, "f"'] & Z \end{tikzcd}$$

is called a *pullback square* if P the pullback of f and g. The $\lrcorner$ us sometimes used to denote that the square is a pullback.

Pullbacks III

Proposition

Let

$$\begin{array}{ccc} P & \xrightarrow{p_2} & B \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle g} \\ A & \xrightarrow[f]{} & Z \end{array}$$

be a pullback square. Then:

- *If g is a monomorphism, then p_1 is a monomorphism;*
- *If g is an isomorphism, then p_1 is an isomorphism;*

The converse is not necessarily true, and the pullback of a epimorphism may not be epic.

Pullbacks IV

Proposition

Consider the following diagram:

$$
\begin{tikzcd}
\bullet \arrow[r] \arrow[d] & \bullet \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \bullet \arrow[d] \\
\bullet \arrow[r] & \bullet \arrow[r] & \bullet
\end{tikzcd}
$$

Assume that the inner right-hand side square is a pullback square, as signalled by the $\lrcorner$.
Then the outer square is a pullback square if and only if the inner left-hand side square is a pullback square.

Pullbacks V

Proposition

A map $X \xrightarrow{f} Y$ is a monomorphism if and only if the following is a pullback square:

$$\begin{tikzcd} X \arrow[r, "1"] \arrow[d, "1"'] & X \arrow[d, "f"] \\ X \arrow[r, "f"'] & Y \end{tikzcd}$$

Limits I

Definition

Let $\mathcal{C}$ be a category and $\mathcal{I}$ a small category. A functor $D : \mathcal{I} \to \mathcal{C}$ is called a *diagram* in $\mathcal{C}$ of shape $\mathcal{I}$.

Definition

Consider a diagram $D : \mathcal{I} \to \mathcal{C}$. A *cone* of the diagram D is an object $A \in \mathcal{C}$ (called *vertex* of the cone) together with a family of morphisms

$$\left(A \xrightarrow{f_I} D(I)\right)_{I \in \mathcal{I}}$$

such that, for every maps $I \xrightarrow{u} J$ in $\mathcal{J}$, the following commutes:

$$\begin{tikzcd}
 & A \arrow[dl, "f_I"'] \arrow[dr, "f_J"] & \\
D(I) \arrow[rr, "D(u)"'] & & D(J)
\end{tikzcd}$$

Limits II

Definition

Let $\mathcal{C}$ be a category and $D : \mathcal{I} \to \mathcal{C}$ a diagram of shape $\mathcal{I}$. A *limit* of D (if it exists), is a cone $\left(L \xrightarrow{p_I} D(I)\right)_{I \in \mathcal{I}}$ with the property that, for any other cone $\left(A \xrightarrow{f_I} D(I)\right)_{I \in \mathcal{I}}$, there exists a unique map $A \xrightarrow{\bar{f}} L$ such that the following commutes for each $I \in \mathcal{I}$

$$\begin{tikzcd} A \arrow[rr, dashed, "\bar{f}"] \arrow[dr, "f_I"'] & & L \arrow[dl, "p_I"] \\ & D(I) & \end{tikzcd}$$

The maps p_I are called the *projections* of the limit and the cone $\left(L \xrightarrow{p_I} D(I)\right)_{I \in \mathcal{I}}$ is usually called a *limiting cone* for the diagram D.

Limits III

Proposition

Let $\mathcal{C}$ be a category and consider a diagram $D : \mathcal{I} \to \mathcal{C}$. Then, if this diagram has a limit in $\mathcal{C}$, that limit is unique up to isomorphism.

Note: As such, if the limit of the diagram $D : \mathcal{I} \to \mathcal{C}$ exists, we denote it by $\varprojlim_{\mathcal{I}} D$ or simply $\varprojlim D$ when $\mathcal{I}$ is implicit by context.

Definition (alternative def. of product)

Let $\mathcal{C}$ be a category and $A, B \in \mathcal{C}$. Let $\mathcal{I}$ be the discrete category with just two points

$$\bullet \qquad \bullet$$

Let $D : \mathcal{I} \to \mathcal{C}$ be the diagram that maps one of the points in $\mathcal{I}$ to A and the other one to B. The limit, if it exists is called the *product* of A and B, denoted as $A \times B$.

Limits IV

Definition

Let $\mathcal{C}$ be a category and I be a set. Consider a set of objects $A_i \in \mathcal{C}$ indexed by I. Let $\mathcal{I}$ be the discrete category whose objects are the elements of I and consider the diagram $D : \mathcal{I} \to \mathcal{C}$ where $D(i) = A_i$. Then the limit $\varprojlim D$, if it exists, is called the product of the $\{A_i\}_{i \in I}$ and is denoted by $\prod_{i \in I} A_i$.

Notes

The product comes equipped with a projection map $\pi_j : \prod_{i \in I} A_i \to A_j$ for each $j \in I$.

Given maps $f_i : C \to A_i$, then there exists a unique map $f : C \to \prod_{i \in I} A_i$ such that $\pi_i f = f_i$.

Limits V

Definition (alternative def. of equalizer)

Let $\mathcal{C}$ be a category and consider two parallel maps $f, g : A \rightrightarrows B$. The *equalizer* of f and g is the limit of the following diagram:

$$\left(\bullet \rightrightarrows \bullet \right) \xrightarrow{D} \left(A \underset{g}{\overset{f}{\rightrightarrows}} B \right)$$

Definition (alternative def. of terminal object)

Let $\mathcal{C}$ be a category and $D : \emptyset \to \mathcal{C}$ the empty diagram. Then a terminal object of $\mathcal{C}$, if it exists, is the limit of D.

Limits VI

Definition (alternative def. of pullback)

Let $\mathcal{C}$ be a category, and consider morphisms

$$
\begin{tikzcd}
& B \arrow[d, "g"] \\
A \arrow[r, "f"'] & Z
\end{tikzcd}
$$

The *pullback* of f and g is the limit of the diagram:

$$
\left(
\begin{tikzcd}
& \bullet \arrow[d] \\
\bullet \arrow[r] & \bullet
\end{tikzcd}
\right)
\xrightarrow{D}
\left(
\begin{tikzcd}
& B \arrow[d, "g"] \\
A \arrow[r, "f"'] & Z
\end{tikzcd}
\right)
$$

Limits VII

Proposition

Let $D : \mathcal{I} \to \mathbf{Set}$ be a diagram. Then, $\varprojlim D$ always exists and is given as a subset of $\prod_{I \in \mathcal{I}} D(I)$ by:

$$\varprojlim D = \{(x_I)_{I \in \mathcal{I}} : x_I \in D(I) \text{ and } Du(x_I) = x_J \text{ for all } I \xrightarrow{u} J \text{ in } \mathcal{I}\}$$

The limiting cone is given by the projections $\varprojlim D \xrightarrow{\pi_I} D(I)$, for each $I \in \mathcal{I}$.

Limits VIII

Theorem

Let $\mathcal{C}$ be a category. Then:

1. *If $\mathcal{C}$ has all products and equalizers, then it has all limits;*
2. *If $\mathcal{C}$ has binary products, terminal object, and equalizers, then $\mathcal{C}$ has finite limits.*
3. *If $\mathcal{C}$ has binary products, then $\mathcal{C}$ has equalizers if and only if $\mathcal{C}$ has pullbacks.*
4. *If $\mathcal{C}$ has pullbacks and terminal object, then $\mathcal{C}$ has finite limits.*